Environmental Modelling & Software 76 (2016) 154–166

Contents lists available at ScienceDirect

Environmental Modelling & Software

journal homepage: www.elsevier.com/locate/envsoft

Robust global sensitivity analysis under deep uncertainty via scenario analysis

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ARTICLE INFO

Article history:
Received 20 March 2015
Received in revised form
30 October 2015
Accepted 7 November 2015
Available online 21 November 2015

Keywords:
Global sensitivity analysis
Robust sensitivity analysis
eFAST
Decision theory
Land use change
Deep uncertainty

ABSTRACT

Complex social-ecological systems models typically need to consider deeply uncertain long run future conditions. The influence of this deep (i.e. incalculable, uncontrollable) uncertainty on model parameter sensitivities needs to be understood and robustly quantified to reliably inform investment in data collection and model refinement. Using a variance-based global sensitivity analysis method (eFAST), we produced comprehensive model diagnostics of a complex social-ecological systems model under deep uncertainty characterised by four global change scenarios. The uncertainty of the outputs, and the influence of input parameters differed substantially between scenarios. We then developed sensitivity indicators that were robust to this deep uncertainty using four criteria from decision theory. The proposed methods can increase our understanding of the effects of deep uncertainty on output uncertainty and parameter sensitivity, and incorporate the decision maker's risk preference into modelling-related activities to obtain greater resilience of decisions to surprise.

1. Introduction

Large, integrated models of complex social-ecological systems are increasingly used to assess the impacts of land use changes and ecosystem services under key influences: future climates, population dynamics, resource scarcity, as well as demand and supply in local and global markets for food, energy, carbon and other commodities (Schaldach et al., 2011; Bateman et al., 2013; Connor et al., 2015). Global sensitivity analysis (GSA) is increasingly being used to analyse these models—identifying critical model input parameters and quantifying the impact of uncertainty in these parameters on model outputs (Saltelli et al., 2008). Knowing influential factors can help to prioritise data collection, identify non-influential model parameters, isolate major parametric uncertainty sources, understand model structure, and check model errors. Sensitivity analysis is often performed on simple models fit to observed data. However, complex social-ecological systems are typically subject to uncertain future conditions, which are usually shaped by local and global drivers, coupled with specific natural, economic, and social trends (Millennium Ecosystem Assessment, 2005a; IPCC, 2007; Moss et al., 2010). Often, probabilities of the occurrence of these future states are not known (or not agreed on by experts) and predictions based on past data are not reliable. This kind of uncertainty is incalculable and uncontrollable, and characterised as *deep uncertainty* (Knight, 1921; Ben-Haim, 2001; Groves and Lempert, 2007; Walker et al., 2010; Wintle et al., 2010; Cox, 2012). Deep uncertainty is distinct from 'probabilistic uncertainty' (Morgan and Henrion, 1992) or 'stochastic uncertainty' (Quade, 1989) which include frequency-based probabilities or subjective (Bayesian) probabilities. Typical sources of deep uncertainty are the future state of the world (objective) and the decision maker (subjective). The presence of deep uncertainty challenges global sensitivity analyses in two main ways. First, we need to understand the influence of deep uncertainty on model parameter sensitivity. Second, we need to quantify sensitivity in a way that is robust to deep uncertainty and can be used to reliably inform investment in gathering and analysing additional data and refining model input parameter estimates.

As traditional predict-then-act approaches are often inadequate for social-ecological assessments when faced with uncontrollable and deep uncertainty (Peterson et al., 2003; Popper et al., 2005), a

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http://dx.doi.org/10.1016/j.envsoft.2015.11.001
1364-8152/

common approach is to identify a set of plausible, internally-consistent future scenarios covering a range of possible conditions (IPCC, 2000; Millennium Ecosystem Assessment, 2005b). When the future is inherently uncertain, using scenario planning rather than a detailed prediction of the most likely future can help decision makers recognize and respond more effectively to changes (Mahmoud et al., 2009; Wilkinson and Kupers, 2013; Kirby et al., 2014). Recent influential efforts using scenarios to address deep uncertainty include the Intergovernmental Panel on Climate Change (IPCC)'s representative concentration pathways (Moss et al., 2010; van Vuuren et al., 2011) and the Millennium Ecosystem Assessment (Millennium Ecosystem Assessment, 2005a). Although scenarios enable a creative, participatory, and systematic approach to exploring potential futures (Peterson et al., 2003; Mahmoud et al., 2009; Polasky et al., 2011; Kirby et al., 2015), their weakness lies in the reliance on a manageably small number of scenarios to capture the full range of future uncertainty (Bryant and Lempert, 2010). In response, recent work in bottom-up decision frameworks (Groves and Lempert, 2007; Bryant and Lempert, 2010; Kasprzyk et al., 2013; Matrosov et al., 2013; Gao et al., 2014; Kwakkel et al., 2014; Herman et al., 2015) has advocated the discovery of many scenarios by applying statistical or data-mining algorithms. Scenarios derived both through traditional participatory means, and more recently through discovery techniques, have been widely used to provide context for systems models in assessing future management plans of natural resources (Bohensky et al., 2006; Bryant and Lempert, 2010; Bryan et al., 2011; Kasprzyk et al., 2013; Davies-Barnard et al., 2014). Sensitivity analysis of these systems models is essential to understand model structure and behaviour, uncertainty, and parameter sensitivity. However, varying individual parameters—a requirement of sensitivity analysis—may jeopardize the internal consistency of scenarios by combining implausible parameter combinations.

One way to quantify the influence of deep uncertainty on model sensitivity is by performing sensitivity analysis (SA) under different scenarios. Two broad SA strategies are commonly used to explore the parameter space: local SA (LSA) and global SA (GSA). GSA overcomes the two main drawbacks of LSA (i.e. exploration only within a small interval around the *baseline* point, and ignorance of input parameter interaction effects) by exploring the full multivariate space of a model simultaneously. Among the several GSA methods proposed, variance-based methods (e.g., Sobol', 1993; Saltelli et al., 2010) decompose the variance of a model output into fractions that can be attributed to model inputs. They have recently been applied in environmental modelling such as hydrology (Nossent et al., 2011; Vezzaro and Mikkelsen, 2012; Shin et al., 2013; Gan et al., 2014), forestry (Miao et al., 2011; Song et al., 2012, 2013), agriculture (Confalonieri et al., 2010; Varella et al., 2010; DeJonge et al., 2012; Zhao et al., 2014), environmental chemistry (Annoni et al., 2011), ecology (Makler-Pick et al., 2011), and waste disposal (Freni and Mannina, 2010; Chen et al., 2012; Cosenza et al., 2013). In particular, the extended Fourier Amplitude Sensitivity Testing (eFAST) has attracted attention (Marino et al., 2008; Chen et al., 2012; Cosenza et al., 2013; Wang et al., 2013; Vanuytrecht et al., 2014; Zhao et al., 2014) due to its computational efficiency (Zhao et al., 2014) and ability to calculate the total sensitivity index (Saltelli et al., 1999).

Determining a sensitivity metric/measure that can reliably identify parameter sensitivities and thereby inform data collection and parameter refinement effort under deeply uncertain future conditions can be framed as a problem of making a robust decision (Polasky et al., 2011)—"*the best possible choice that one found by eliminating all the uncertainty possible within available resources, and then choosing, with known and acceptable levels of satisfaction and risk*" (Ullman, 2006). Decision theory offers scientifically-sound and practically-feasible strategies to make optimal decisions under uncertainty (Morgan et al., 1992). In situations of deep uncertainty, a combination of scenario planning and decision theory can be a useful way to expand the space of possible states and associated outcomes, and to make decisions given current understanding. A strategy in decision theory usually characterises a decision (in our case, quantifying parameter sensitivities) as a function of different possible states of the world (scenarios), alternatives/choices, utility functions between scenarios and alternatives, and a decision criterion. Then, a good decision is made in situations of deep uncertainty by incorporating the decision maker's risk preference over different outcomes in the decision criterion. The robustness of this decision is provided by the consideration of multiple possible futures with acceptable levels of satisfaction and risk.

We undertook a GSA for a complex social-ecological systems model—the Australian continental Land Use Trade-Offs (LUTO) model (Bryan et al., 2014; Connor et al., 2015; Bryan et al., in review), and assessed the impact of scenario choice on parameter sensitivities. LUTO provides an integrated, evidence-based assessment of a range of possible outlooks for Australian land use and ecosystem services (Bryan et al., 2014; Connor et al., 2015). The model uses scenarios combining both global and domestic drivers to explore the implications of uncertain future conditions (Hatfield-Dodds et al., 2015). It is necessary to understand how the scenarios influence model parameter sensitivities and to identify a parameter sensitivity metric that is robust under different scenarios. We applied eFAST in a GSA of the LUTO model under four global outlooks. We then statistically evaluated the influence of different scenarios on output uncertainty resulting from the variation in input parameters. We identified how the scenarios affected influential, non-influential, and interacting parameters in the model. We also tested for statistical differences in the variance contribution of input parameters to model outputs under different scenarios. Finally, we calculated a set of sensitivity indicators that were robust to uncertain future conditions by using four criteria from decision theory. We discuss the implications for undertaking GSA on complex models under conditions of deep uncertainty.

2. The social-ecological model

2.1. Model overview and scenarios

The LUTO model identifies potential land use futures for Australia's intensively managed agricultural land—an area of 85.3 million ha stretching from central and eastern Queensland, through New South Wales, Victoria and South Australia, to the cropping land of south-west Western Australia (Fig. 1). Within the study area, 33 million ha are currently used for cattle production, 18 million ha for sheep production, 3 million ha for dairy, and 25 million ha for grain production. The model runs on an annual time step over the time period 2013–2050, at a spatial resolution of 1.1 km (0.01°) grid cells. It models economic profitability and competition for land between five main land use classes: agriculture, carbon plantings, environmental plantings, biofuels, and bioenergy. The LUTO model integrates a range of environmental and economic data layers and models of the production, price, and costs of each land use over time, as well as the impacts of each land use for a range of ecosystem services including carbon sequestration, biodiversity, food production, water, and energy. Then by aggregating the land use decisions made in each grid cell, the LUTO model identifies potential land use changes over time and the resultant ecosystem service impacts of land use futures. Details of the LUTO model are described in Connor et al. (2015) and Bryan et al. (2014).

Global context for the LUTO model was provided by four global outlooks (termed L1, M3, M2, and H3)—defined as plausible,

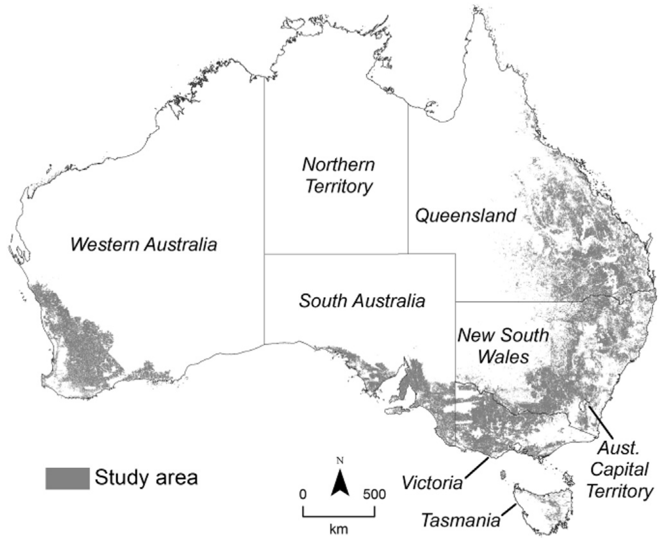

Fig. 1. The study area in Australia.

internally consistent scenarios for the global economy, population, greenhouse gas emissions, and climate (Hatfield-Dodds et al., 2015). These scenarios were developed based on three cumulative greenhouse emissions trajectories and three UN population projections. They are not intended to be comprehensive or cover all possible future outcomes, and no probability or likelihood is associated with them. L1 is characterised by very strong global action on climate change, low global population, and the lowest degree of climate change. Both M2 and M3 are mid-range climate outlooks and differ insofar as M3 has higher population and strong emissions abatement effort while M2 has a lower population with modest abatement effort. H3 involves no emissions abatement, high population, and severe climate change (Table 1). Total demand for food and energy varies widely due to differences in population and average global income. The modelled grain, carbon, oil, and livestock price paths under the four scenarios are also summarised in Table 1.

2.2. Input factors and output variables

We selected 50 parameters classified into 9 groups for the sensitivity and uncertainty analysis of the LUTO model (Table 2). For spatial input parameters, we used a single factor approach such that all grid cells were varied by this factor. Similarly, a multiplier was also applied to time-series values in the scenario parameter group. Because of the absence of information about the prior probability distributions for each parameter, a uniform distribution was assumed. For parameters lacking information determining its range, we set bounds of 30% each side of the reference value following Song et al. (2012) and Van Oijen et al. (2005).

We analysed the sensitivity and uncertainty of 24 model outputs (as shown in Table 3) in 2050 from the LUTO simulation to variation in each of the input parameters (Table 2).

3. Methods

We conducted a number of experiments to determine whether the uncertainty and sensitivity analysis results were different depending on scenario, and to build a single set of indicators that are robust to future uncertainty.

3.1. Global sensitivity analysis using eFAST

eFAST is a variance decomposition method, which uses a periodic sampling method and a Fourier transformation to partition the whole variance of the model output and quantify the degree to which variation in each input factor accounts for the output variance. A periodic sampling approach is used to generate a search curve in the parameter space and partitioning is implemented by assigning the periodic sample of each parameter with a distinct frequency. Then a Fourier transformation is applied to the model output to measure how strongly a factor's frequency propagates from the input to the output, i.e., the variance contribution of the factor to the whole variance of the output (Saltelli et al., 1999; Marino et al., 2008). The calculation process of first-order and total sensitivity indices using the eFAST method is shown in Supplementary Material A.

We applied eFAST in uncertainty and sensitivity analyses of the LUTO model. LUTO was written in the Python programming language (van Rossum and The Python Community, 2013). With each model run taking about 40 h, it is relatively computationally intensive and this poses a computing challenge for GSAs which usually require many model simulations. To address this computing challenge we used a reduced resolution version (3.3 km grid cells)

Table 1
Dimensions of the four global outlooks assessed.

Indicator	Unit	Value in 2010	Scenario			
			L1	M3	M2	H3
Population and economic growth						
Population in 2050	Billion people	6.9	8.1	10.6	9.3	10.6
World GDP per capita in 2050	US$ '000 2010 cap^{-1}	8.8	20.0	18.6	19.3	18.6
World GDP in 2050	US$ '000 billion	61.0	161.6	197.0	179.1	197.8
Climate						
Temperature increase by 2100	°C	–	1.3–1.9	2.0–3.0	2.0–3.0	4.0–6.1
Emissions and abatement						
Benchmark Raditative Concentration Pathway (RCP)	–	–	RCP3-PD	RCP4.5	RCP4.5	RCP8.5
Radiative forcing in 2100	Wm^{-2}	–	2.6	4.5	4.5	8.5
Atmospheric emissions in 2100	CO_2 ppm	–	445 (declining)	650 (stable)	650 (stable)	1360 (rising)
Global abatement effort		–	Very strong	Strong	Modest	No action
Emissions per person in 2050	$tCO_2\ cap^{-1}$	7.0	3.1	4.3	5.0	8.7
Coverage of abatement policy	–	–	All sources	All, excluding emissions from livestock		No sources
Projected prices						
Grain price increase from 2010 to 2050	AUD$	–	75%	118%	11%	61%
Carbon price (global) in 2050	AUD$	–	200	119	59	0
Livestock price increase from 2010 to 2050		–	147%	112%	22%	61%
Oil price increase from 2010 to 2050		–	42%	44%	45%	43%

Table 2
Input parameters of the LUTO model used in the sensitivity and uncertainty analyses.

Parameter group	Parameter	Description	Unit	Spatial layer	Value range	
					Min	Max
Scenario	CropPricePath	Crops and horticultural price path	n/a	N		
	LivestockPricePath	Livestock price path	n/a	N		
	CarbonPricePath	Carbon price path	$ tCO_2^{-1}	N		
	OilPricePath	Oil price path	n/a	N		
	ElectricityPricePath	Average wholesale electricity price	AUD MWh^{-1}	N		
	AgriculturalProductivity	Productivity increases for crops, beef, sheep, dairy and horticulture	n/a	N	0	3
	TreeGrowthProductivity	Productivity increases for trees	n/a	N	0	1
	AdoptionHurdleRate	Economic adoption hurdle rate. Agricultural returns increased by hurdle rate	n/a	N	1	5
	BiodiversityLevy	Amount of levy on carbon plantings	%	N	0	30
	BiodiversityFund	Initial biodiversity fund amount	$ yr^{-1}	N		
	DiscountRate	Discount rate	n/a	N		
Agriculture profit function	CommodityPricePrimary	Farm gate price for primary product	$ ha^{-1}	Y		
	CommodityPriceSecondary	Farm gate price for secondary product (milk and wool)	$ ha^{-1}	Y		
	AgriculturalYield	Agriculture product quantity produced	t ha^{-1} or DSE ha^{-1} (dry sheep equivalents)	Y		
	LivestockDisposals	Livestock disposals	DSE ha^{-1}	Y		
	AgAreaCosts	Area cost	$ ha^{-1}	Y		
	AgQuantityCosts	Quantity cost = actual QC* Q	$ ha^{-1}	Y		
	AgFixedOperatingCosts	Fixed operating cost	$ ha^{-1}	Y		
	AgFixedLabourCosts	Fixed labour cost	$ ha^{-1}	Y		
	AgFixedDepreciationCosts	Fixed deprication cost	$ ha^{-1}	Y		
	AgWaterCosts	Water delivery cost	$ ha^{-1}	Y		
Tree inputs	CarbSeqRateCP	Carbon plantings average annual CO2 sequested over 20 yrs	$tCO_2\ ha^{-1}\ yr^{-1}$	Y		
	CarbSeqRateEP	Environmental plantings average annual CO2 sequested over 20 yrs	$tCO_2\ ha^{-1}\ yr^{-1}$	Y		
	RiskFireDroughtCP	Percentage of CO2 sequestration to use as 100 year accumulated biomass	n/a	Y		
	RiskFireDroughtEP	Percentage of CO2 sequestration to use as 100 year accumulated biomass	n/a	Y		
	CarbSeqRiskBuffer	Risk buffer for carbon monoculture and environmental plantings	n/a	N		
Tree costs	EstabCostCP	Initial cost of establishing carbon plantings	$ ha^{-1}	Y		
	EstabCostEP	Initial cost of establishing environmental plantings	$ ha^{-1}	Y		
	EstabCostWP	Initial cost of establishing biomass plantings	$ ha^{-1}	Y		
	AnnMangCostCP	Annual management costs for carbon plantings	$ $ha^{-1}\ yr^{-1}$	N		
	AnnMangCostEP	Annual management costs for environmental plantings	$ $ha^{-1}\ yr^{-1}$	N		
	AnnMangCostWP	Annual management costs for biomass plantings	$ $ha^{-1}\ yr^{-1}$	N		
	HarvestCostWP	Harvest cost for biomass plantings. Every 10 years.	$ ha^{-1}	N		
Biofuel inputs	BiomassGrowthRateWP	Woody perennials plantings	t $ha^{-1}\ decade^{-1}$	Y		
	WheatStubbleYield	Biofuels wheat stubble	t $ha^{-1}\ yr^{-1}$	Y		
	WheatGrainYield	Biofuels wheat yield (grain)	t $ha^{-1}\ yr^{-1}$	Y		
Biofuel costs	WheatPrice	Farm gate price for primary product. Price of winter cereals when grain sold for food and stubble for biofuel.	$ ha^{-1}	Y		
	WheatAreaCost	Area cost	$ ha^{-1}	Y		
	WheatQuantityCost	Quantity cost = actual QC* Q	$ ha^{-1}	Y		
	WheatFixedOperatingCost	Fixed operating cost	$ ha^{-1}	Y		
	WheatFixedLabourCost	Fixed labour cost	$ ha^{-1}	Y		
	WheatFixedDepreciationCost	Fixed deprication cost	$ ha^{-1}	N		
Climate impacts	CCImpactDrylandCrops	Climate impacts for dryland crops	n/a	Y		
	CCImpactDrylandLivestock	Climate impacts for dryland crops	n/a	Y		
	CCImpactTreeGrowth	Climate impacts for trees	n/a	Y		
Water	WaterInterceptionTrees	Water impact as rate of tree interception	ML $ha^{-1}\ year^{-1}$	Y		
	WaterLicensePrice	Price of water intercepted by trees	$ ML^{-1}	Y		
	CCImpactWaterLicensePrice	Cell value of climate impact on water run-off used to adjust water price	%	Y		
Others	AgPriceElasticityOfDemand	Elasticity of demand		N	−0.8	−0.4
	BioenergyProfitShare	Proportion of difference between petrol price and biofuel retail price paid to producer		N		

Table 3
A list of the major LUTO output variables used in the analyses.

Output variable	Description	Unit
ProductionGrains	Total amount of grains produced in a year	tonne
ProductionOtherCrops	Total amount of other crops produced in a year	Tonne
ProductionDairy	Total number of dairy cattle	head
ProductionBeef	Total number of beef cattle	head
ProductionSheep	Total number of sheep	head
CarbonSequestration	Creditable carbon sequestration	tCO2e
WaterIntercepted	Annual water interception from trees	ML
BiodiversityServices	Area weighted measure of biodiversity value	%
AgriculturalProduction	Value of agricultural production in 2010 prices	$
EnergyBiofuel	Total biofuels produced	ML
EnergyElectricity	Total bio-electricity produced	GJ
AreaGrains	Total area of grains	ha
AreaOtherCrops	Total area of other crops	ha
AreaDairy	Total area of dairy	ha
AreaBeef	Total area of beef	ha
AreaSheep	Total area of sheep	ha
AreaCarbonPlantings	Total area of carbon plantings	ha
AreaEnvironmentalPlantingsC	Total area of environmental plantings (carbon price)	ha
AreaEnvironmentalPlantingsCBF	Total area of environmental plantings (carbon price and carbon price + biodiversity fund)	ha
AreaWheat (Food + Biofuel)	Total area of crops planted to wheat for food and biofuels	ha
AreaWheat (Biofuel)	Total area of crops planted to wheat for biofuels	ha
AreaWoodyPerennials (Biofuel)	Total area of woody perennials for biofuels	ha
AreaWheat (Food + Electricity)	Total area of crops planted to wheat for food and electricity	ha
AreaWoodyPerennials (Electricity)	Total area of woody perennials for electricity	ha

and employed parallel programming techniques in the sensitivity and uncertainty analyses (Bryan, 2013; Zhao et al., 2014). Parameter set generation was undertaken using Matlab R2013b (The MathWorks Inc., USA). For each parameter set, an individual LUTO simulation was parameterised, and a compute job was created and submitted to CSIRO's computing clusters, and executed in parallel. The sensitivity and uncertainty analyses were also conducted using Matlab R2013b (The MathWorks Inc., USA).

The eFAST sampling procedure implements a sinusoidal function of a particular frequency ω_i associated with the input parameter x_i. The frequency assigned to the parameter is based on the sample from 1 to the number of samples per search curve, N_s. Saltelli et al. (1999) suggested the minimum value for N_s is 65, and optimal values of ω_i are between $16{\cdot}N_r$ and $(16{\cdot}N_r + 48{\cdot}N_r)$, where N_r is the number of resamples. We used 145 samples per search curve without resampling (i.e. $N_s = 145, N_r = 1$) where ω_i was calculated as 18. Thus, the total number of model simulations was 7250 for 50 parameters (i.e. 145×50) for each scenario. The required number of model simulations was confirmed by a convergence test, which incrementally increased the number of model runs and quantified the output difference between two subsequent increments (Song et al., 2013; Cosenza et al., 2014). The simulation results were used for both uncertainty and sensitivity analyses.

We identified influential input parameters by calculating both the first-order sensitivity index S_i and total sensitivity index S_{Ti}. In line with Cosenza et al. (2014), for each output variable, a threshold of $S_i > 0.01$ and $S_{Ti} > 0.1$ was imposed to identify influential parameters. We displayed the results of the eFAST sensitivity analyses in grid diagrams (Song et al., 2012).

3.2. Uncertainty analysis and between-scenario differences in model outputs

Uncertainty analysis aims to depict the entire set of possible model outcomes, together with their associated frequencies of occurrence (Loucks et al., 2005). We conducted uncertainty analyses for investigating the between-scenario differences of outcome uncertainty derived from the variation in input parameters with eFAST. First, we calculated coefficients of variation (CV) of model outputs in order to justify our use of the variance decomposition method and GSA work as very small CV values indicate that understanding the contribution of model inputs to model output variance would be of little value (Cosenza et al., 2014). CV was also used to describe the degree of dispersion and shape of output frequency distributions under different scenarios. Second, to test for significant differences between the outputs from different scenarios, the non-parametric Kruskal–Wallis test (Corder and Foreman, 2009) was used due to the lack of normality in the data. The level of significance was $\alpha = 0.05$ in both tests. Last, we visualised between-scenario differences in uncertainty using a combined violin and box plot.

3.3. Correlation analyses of between-scenario parameter sensitivity

Using eFAST in a GSA, we quantified the sensitivity of model outputs to uncertainty in model input parameters under the four scenarios. To quantify the differences between sensitivity indices under different scenarios, we used the non-parametric Spearman's rank correlation test which does not require normality in the data (Jackson, 2012). The correlation coefficient describes the strength of association between sensitivity index sets. Following Walker and Almond (2010), we defined the relationship as strong when an ρ value varies between ±(0.8~1), as moderate when ρ varies between ±(0.6~0.79), as weak when ρ varies between ±(0.4~0.59), and as little or no association when ρ varies between ±(0.0~0.39).

3.4. Robust sensitivity indicators

A robust sensitivity indicator ($S_{RTi}^{criterion}$) of each input parameter (x_i) for each model output was calculated based on the total sensitivity indices ($S_{Ti,L1}$, $S_{Ti,M3}$, $S_{Ti,M2}$, and $S_{Ti,H3}$) for the four scenarios. We used four criteria from decision theory (maximax, weighted average, minimax regret, and limited degree of confidence) to calculate robust sensitivity indicators.

(1) Maximax

The maximax criterion (Anderson et al., 2012) is an aggressive decision making rule under conditions of uncertainty. The decision criterion finds the maximum possible payoff (profit or loss) for each alternative, and then selects the alternative with the greatest maximum payoff. In decision theory, payoffs are the outcomes for any combination of an alternative *a* (scenario *a*) and state of nature *b* (scenario *b* that the future will follow). Here, payoff $p_{a,\ b}$ is defined as the total index difference between two scenarios (*a* and *b*). Therefore, for an input parameter x_i, payoff $p_{a,\ b}$ is defined as:

$$p_{a,b} = S_{Ti,a} - S_{Ti,b} \tag{1}$$

The payoffs associated with all possible combinations of alternatives and states of nature constitute a payoff table. The maximum possible payoff for each alternative is selected first, and then the alternative *j* with the greatest maximum payoff is identified, such that:

$$\max_a\left(\max_b\left(p_{a,b}\right)\right) \rightarrow j \tag{2}$$

where → is a selection symbol that identifies the final alternative scenario *j* according to the identification condition of $\max_a(\max_b(p_{a,b}))$.

Thus, the total index $S_{Ti,j}$ under scenario *j* is the robust sensitivity indicator S_{RTi}^{max} for the input parameter x_i ($i = 1, 2,\ldots,k$), as shown below:

$$S_{RTi}^{max} = S_{Ti,j} \tag{3}$$

In our case, for a model output, the maximax criterion ensures that no parameter that is influential in any scenario is omitted. Here, the effect of applying this criterion to an input parameter is equivalent to selecting the maximum total sensitivity index over all four scenarios.

(2) Weighted average

We used a weighted average criterion to calculate the robust sensitivity indicator S_{RTi}^{avg} by aggregating the four total sensitivity indices ($S_{Ti,L1}$, $S_{Ti,M3}$, $S_{Ti,M2}$, and $S_{Ti,H3}$), in order to identify the input parameters most influential over all four scenarios. S_{RTi}^{avg} is the sum of the product of the weight β_a of each scenario *a* and the total sensitivity index $S_{Ti,\ a}$ under this scenario.

$$S_{RTi}^{avg} = \sum_a \left(\beta_a \cdot S_{Ti,\ a}\right) \tag{4}$$

where $a = L1, M3, M2, and\ H3$. As we have no information on occurrence probability, we specified equal weights for all four scenarios.

(3) Minimax regret

The minimax regret criterion (Anderson et al., 2012) was used to minimise the regret of accepting sensitivity indices from a scenario which does not eventuate. In this criterion, the regret (opportunity loss) is used rather than payoff. Here, for each state of nature *b* (scenario *b*), the regret $r_{a,\ b}$ is defined as the absolute difference between the total index $S_{Ti,\ a}$ in an alternative scenario *a* and $S_{Ti,\ b}$.

$$r_{a,b} = \left|S_{Ti,a} - S_{Ti,b}\right| \tag{5}$$

The regret values from all possible combinations of alternative scenarios and states of nature can form a regret table, where the maximum regret value for each alternative scenario can be identified as $\max_b(r_{a,b})$ and then the alternative scenario *j* with the lowest maximum regret value is selected.

$$\min_a\left(\max_b\left(r_{a,b}\right)\right) \rightarrow j \tag{6}$$

where → is a selection symbol that identifies the final alternative scenario *j* according to the identification condition of $\min_a(\max_b(r_{a,b}))$.

Finally, the total sensitivity index $S_{Ti,\ j}$ under the scenario *j* was selected to represent the robust sensitivity indicator S_{RTi}^{reg} for each input parameter x_i ($i = 1, 2,\ldots,k$) for unknown future conditions.

$$S_{RTi}^{reg} = S_{Ti,j} \tag{7}$$

(4) Limited degree of confidence (LDC)

The LDC criterion (Lempert and Collins, 2007; McInerney et al., 2012) was used to balance the discounted sum of total sensitivity indices S_{RTi}^{avg} and the minimax regret performance. The criterion was built on the decision-maker only having a limited degree of confidence in the best estimate probability function of the occurrence of the scenarios. The robust sensitivity indicator S_{RTi}^{ldc} can be represented as:

$$S_{RTi}^{ldc} = \gamma \cdot S_{RTi}^{avg} + (1-\gamma) \cdot S_{RTi}^{reg} \tag{8}$$

where γ lies between 0 and 1, and represents the decision-maker's degree-of-confidence in the underlying probability distributions. When $\gamma = 0$, this criterion is equivalent to the minimax regret and when $\gamma = 1$ it is equivalent to the weighted average criterion. We set $\gamma = 0.5$ such that the four scenarios have equal occurrence probability.

4. Results

Of the 24 output variables, the five most important ones (*AgriculturalProduction*, *CarbonSequestration*, *WaterIntercepted*, *BiodiversityServices*, and *EnergyBiofuel*) were selected to illustrate how the sensitivity of these outputs to sampled parameters behaved differently under the four scenarios, and how new robust indicator sets were developed. The full set of results is presented in the Supplementary Material Figs. B1–B2, Table B1, and Figs C1–C9.

4.1. Between-scenario differences in model outputs

Out of the CV values of the 24 outputs under the four scenarios, only the outputs *AreaOtherCrops* and *AreaDairy* had relatively low CV values: approximately 0.04 and 0.1, respectively, across the four scenarios (Supplementary Material Fig. B1). The CV values in the other output variables exceeded 0.2 under the four outlooks. Thus, for most outputs, there were substantial differences in the CV values of different scenarios, thereby affirming the need for sensitivity analysis of model outputs.

Between-scenario uncertainty in the 24 outputs was significant under the specified variation in input parameters (Supplementary Material Fig. B2). Some outputs such as *ProductionBeef* (Supplementary Material Fig. B2 (d)) and *AgriculturalProduction* (Fig. 2) had a more normal distribution, while the distributions of others such as *EnergyBiofuel* (Supplementary Material Fig. B2(j) or Fig. 2) and *AreaGrains* (see Supplementary Material Fig. B2(1)) were more skewed. Although these distributions appeared similar under the four scenarios, they differed in magnitude. For example, the most common values for *ProductionBeef* were approximately calculated as 12.32, 15.49, 17.72, and 20.03 million head for L1, M3,

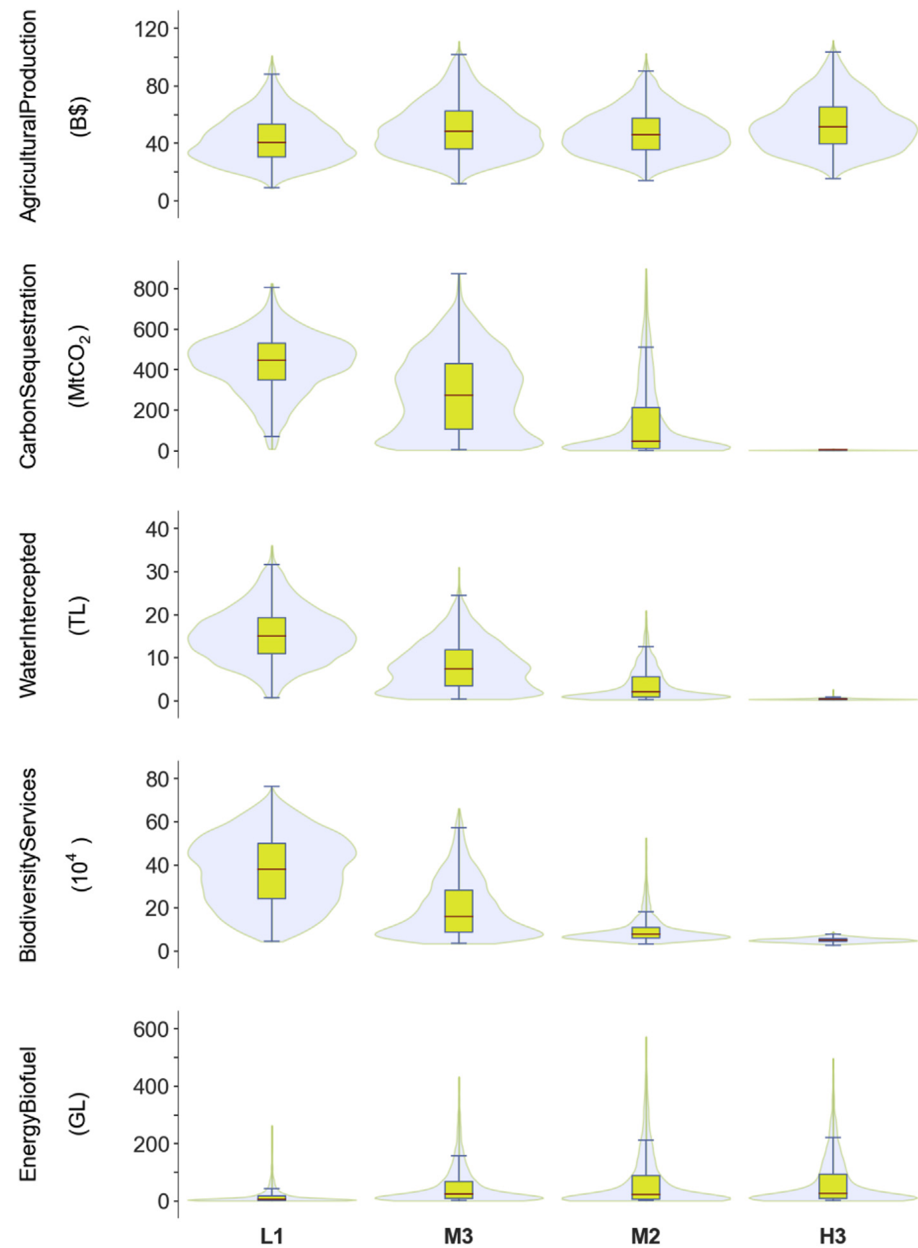

Fig. 2. Empirical frequency distributions for the five main outputs under four scenarios. The line red in the box is the median; the edges of the box are the lower hinge (the 25th percentile, Q1) and the upper hinge (the 75th percentile, Q3), and the whiskers extend to 1.5 × (Q3 − Q1). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

M2, and H3, respectively. Significant between-scenario differences for 23 of the model 24 outputs (all except *AreaWoodyPerennials(Electricity)*) were identified by the Kruskal–Wallis test (Supplementary Material Table B1). Therefore, ignoring scenario impacts on output uncertainty in a typical GSA could result in a biased and unrepresentative view of model structure, performance, parameter sensitivity, and output uncertainty.

4.2. Sensitivity of outputs to input parameters under different scenarios

Supplementary Material Fig. C1–C8 reported the full results of first-order and total sensitivity effects for all 50 input parameters on the 24 output variables, under the four scenarios. Different numbers of influential parameters (i.e. where $S_i > 0.01$ and $S_{T\,i} > 0.1$) were found under the four scenarios. There were 33, 39, 46, and 43 influential parameters under L1, M3, M2, and H3, respectively, for all outputs. The differences of the main and total sensitivity effects of the ten (or less) most influential input parameters for the 24 outputs are visualised in Supplementary Material Fig. C9(a–x).

Substantial differences were found between scenarios in the total sensitivity effects (S_{Ti}) reflecting the influence of input parameters on the five main outputs (Fig. 3 and Fig. 4). For the output *AgriculturalProduction*, the second most influential parameter varied between scenarios: *CommodityPricePrimary* for scenario L1, *WheatGrainYield* for scenarios M3 and H3, and *AgriculturalYield* for H3. *AgriculturalProductivity*, *WheatGrainYield*, *AgriculturalYield*, and *CommodityPricePrimary* were the top four most influential parameters for the four scenarios, except *WheatGrainYield* was non-influential under L1.

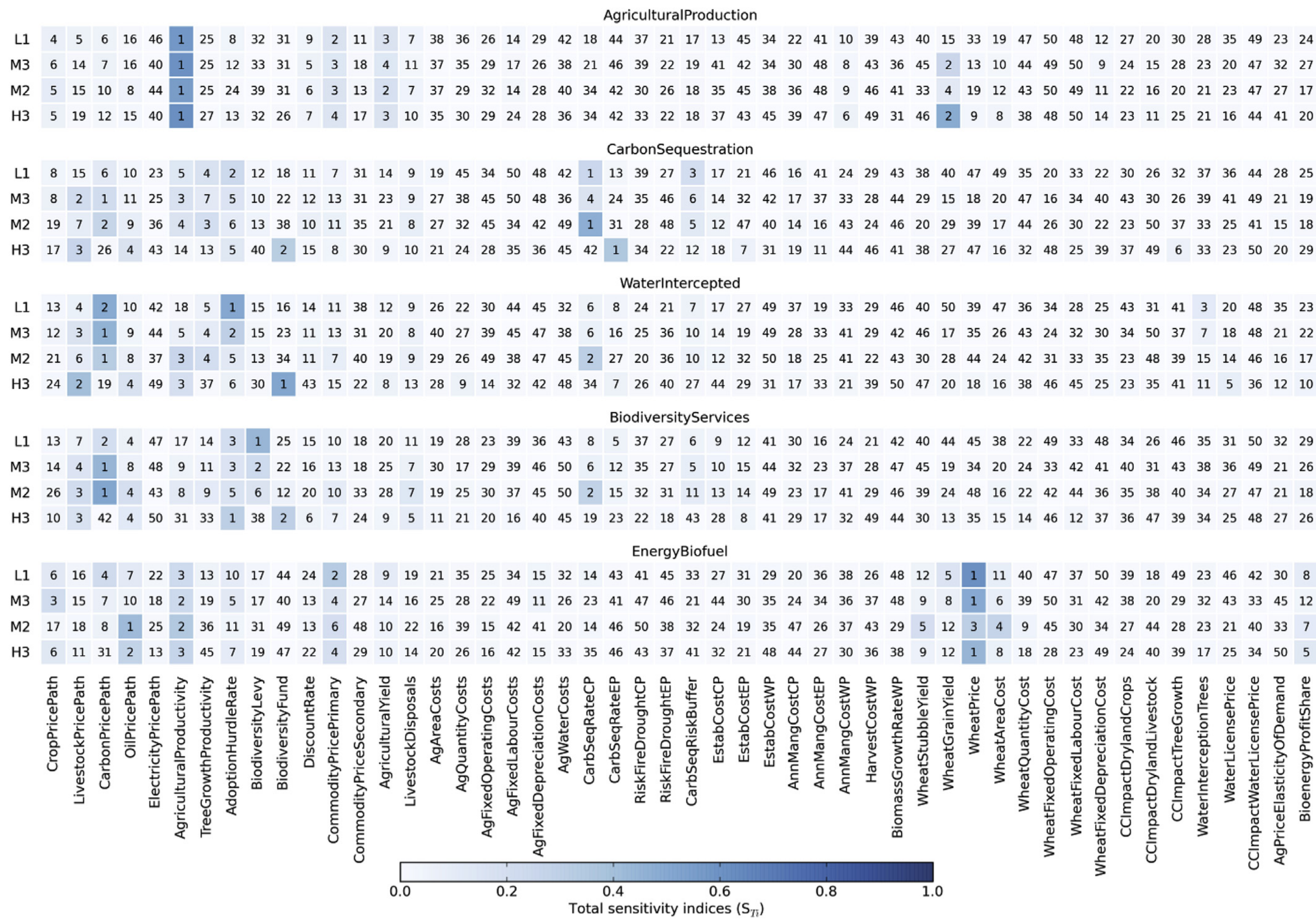

Fig. 3. Total sensitivity effects of 50 input parameters on the five outputs under the four scenarios. Colours in the grid cells represent the total sensitivity effects and numbers represent their rankings of influence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

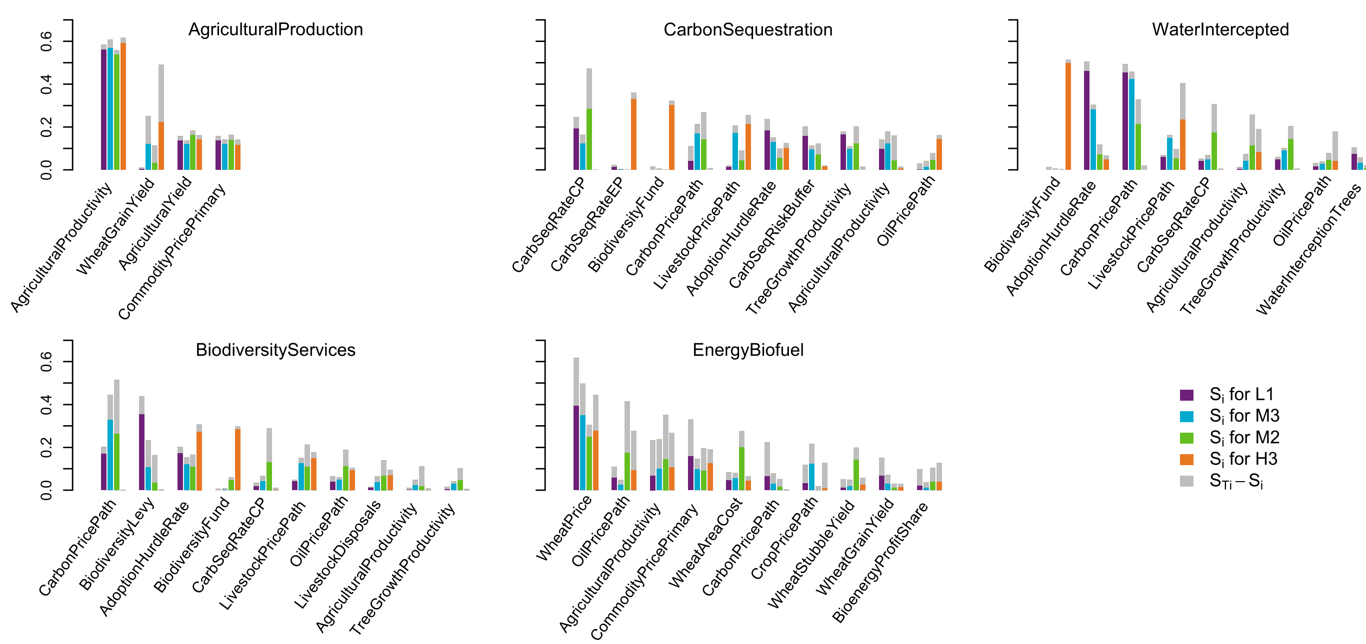

Fig. 4. S_i and S_{Ti} for the influential input parameters for the five outputs. The parameters are ranked based on the maximum S_{Ti} across the four scenarios.

For *CarbonSequestration*, ten input parameters were influential, but the top three under the four scenarios included nine different parameters (*CarbSeqRateCP*, *AdoptionHurdleRate*, and *CarbSeqRiskBuffer* in L1; *CarbonPricePath*, *LivestockPricePath*, and *AgriculturalProductivity* in M3; *CarbSeqRateCP*, *CarbonPricePath*, and *TreeGrowthProductivity* in M2; and *CarbSeqRateEP*, *BiodiversityFund*, and *LivestockPricePath* in H3).

Nine parameters were influential for *WaterIntercepted*. Under L1, *AdoptionHurdleRate*, *CarbonPricePath*, *WaterInterceptionTrees* were influential. *CarbonPricePath* was the most important input under both M3 and M2. For M3, the influential parameters still included *AdoptionHurdleRate*, *LivestockPricePath*, and *TreeGrowthProductivity*. For M2, the second most important parameter was *CarbSeqRateCP*, followed by *AgriculturalProductivity*, *TreeGrowthProductivity*, and *AdoptionHurdleRate*. *BiodiversityFund* was influential only in scenario H3.

Similarly, input parameters influencing *BiodiversityServices* were heavily dependent upon the scenario. *BiodiversityLevy*, *CarbonPricePath*, and *AdoptionHurdleRate* significantly affected the output under both scenarios L1 and M3. However, the ranking of these three parameters and their magnitudes of total effects were quite different. More parameters were identified as influential under M2. Under H3, the effect of *BiodiversityLevy* was negligible, but *AdoptionHurdleRate*, *BiodiversityFund*, *LivestockPricePath*, and *OilPricePath* were important.

For *EnergyBiofuel*, seven influential parameters were identified under L1, with *WheatPrice*, *CommodityPricePrimary*, and *AgriculturalProductivity* the top three. Under M3, five parameters were influential (*WheatPrice*, *AgriculturalProductivity*, and *CropPricePath* held the top three positions). Seven parameters were significant under M2 (*OilPricePath*, *AgriculturalProductivity*, and *WheatPrice* ranked top three). Six parameters were influential under H3, with *WheatPrice*, *OilPricePath*, and *AgriculturalProductivity* the top three.

Correlations in the sensitivity of outputs to model inputs between scenarios were varied. For *CarbonSequestration*, *WaterIntercepted*, and *BiodiversityServices*, there was a weak relationship between the sensitivity index pairs involving H3 (Table 4). 60% of all scenario pairings for the five main model outputs showed moderate to weak correlation in the influence of input parameters.

4.3. Robust sensitivity analyses

Given the difference in input parameter influence and model sensitivities demonstrated above, we present four sets of robust sensitivity indicators by applying four decision criteria. For *AgriculturalProduction*, the four decision criteria resulted in the same parameter ranking in the sensitivity indicators: *AgriculturalProductivity*, *WheatGrainYield*, *AgriculturalYield*, and *CommodityPricePrimary* (Fig. 5).

Variation was also apparent for the other outputs. In particular, parameters whose sensitivity index was only high in one scenario, could be identified as influential by the maximax criterion, but is likely to be non-influential under other decision criteria, especially under the minimax regret criterion whose purpose is to reduce the worst-case performance (i.e. selecting the wrong index that is not robust to the four scenarios). For instance, the sensitivity indices for *CarbSeqRateEP* to the output *CarbonSequestration*, and *BiodiversityFund* to the outputs *CarbonSequestration*, *WaterIntercepted*, as well as *BiodiversityServices* were significant under H3 but insignificant under the other scenarios (Fig. 5). These parameters were influential when the decision maker wants to identify all parameters influential under any scenario (by applying the maximax criterion). However, the regrets (opportunity loss) were large when the future does not follow the trajectories of H3. Hence, a risk-averse decision maker may regard these parameters as non-influential (i.e. by applying the minimax regret criterion).

The weighted average and the minimax regret criterion generally identified similar influential parameters, but differed in the ranking of these parameters. This is because the two criteria differ in the aggregation method of total sensitivity indices over all scenarios. The former focuses on identifying influential parameters over all scenarios while the latter aims for minimising the regret of selecting the wrong alternative. For example, for the output *BiodiversityServices*, *CarbonPricePath* had higher total sensitivity indices over the four scenarios than *BiodiversityLevy*. Under M3 and M2, *CarbonPricePath*'s S_{Ti} was >40% and *BiodiversityLevy*'s S_{Ti} was ≈20% (Fig. 5). Therefore, *CarbonPricePath* and *BiodiversityLevy* ranked the first and second, respectively, when the weighted average criterion was applied, but this ranking was reversed under the minimax regret criterion due to larger opportunity loss in case of selecting a wrong alternative for parameter *CarbonPricePath*. In our case, the LDC criterion balanced the effects of the weighted average and the minimax regret criteria.

Table 4
Correlation coefficients of between-scenario sensitivity indices for the five main outputs.

Model output	Scenario pair					
	L1~M3	L1~M2	L1~H3	M3~M2	M3~H3	M2~H3
AgriculturalProduction	0.89	0.88	0.81	0.94	0.95	0.94
CarbonSequestration	0.78	0.67	**0.47**	0.78	**0.43**	**0.43**
WaterIntercepted	0.81	0.75	**0.42**	0.90	**0.51**	**0.42**
BiodiversityServices	0.87	0.84	**0.46**	0.92	**0.49**	**0.53**
EnergyBiofuel	0.89	0.63	0.66	0.66	0.72	0.77

Note: the weak correlation relationships are marked in bold.

5. Discussion

5.1. Influence of deep uncertainty on output uncertainty and parameter sensitivity

The future state of the world (such as future global economy, population, greenhouse gas emissions, and climate) is characterized by deep uncertainty for which "there is no objective measure of the probability" (Knight, 1921; Ben-Haim, 2001). We performed global sensitivity and uncertainty analyses of the LUTO model—a large, integrated social-ecological system model of Australian land use futures, under deep uncertainty. The type of uncertainty surrounding future drivers is probabilistically intractable and we considered it using internally consistent scenarios, each of which is a structured account of a plausible climate, population, carbon, energy, and economic future. We used eFAST which has proven to be amongst the most reliable and efficient global sensitivity analysis methods. Varying the scenario parameters as in typical GSA (Sobol', 1993; Saltelli et al., 1999) was not appropriate as it violated internal consistency and risked running the model under implausible or impossible conditions. GSA undertaken for each scenario found significant differences in the uncertainty of outputs, and in the influence of model inputs.

We found that model outputs resulting from variation in model inputs in GSA were significantly different under the four scenarios as indicated by the CV test and Kruskal–Wallis test. For some outputs, different scenarios even led to different distributional shapes. Scenarios captured key ingredients of deep uncertainty about future drivers and the LUTO model responded in a strongly non-linear way to these drivers. This could be seen in some outputs where the impact of a parameter's interactions with other parameters on the outputs ($S_{Ti} - S_i$) was much greater than that of the parameter's own contribution (S_i). Different scenarios provided different drivers of

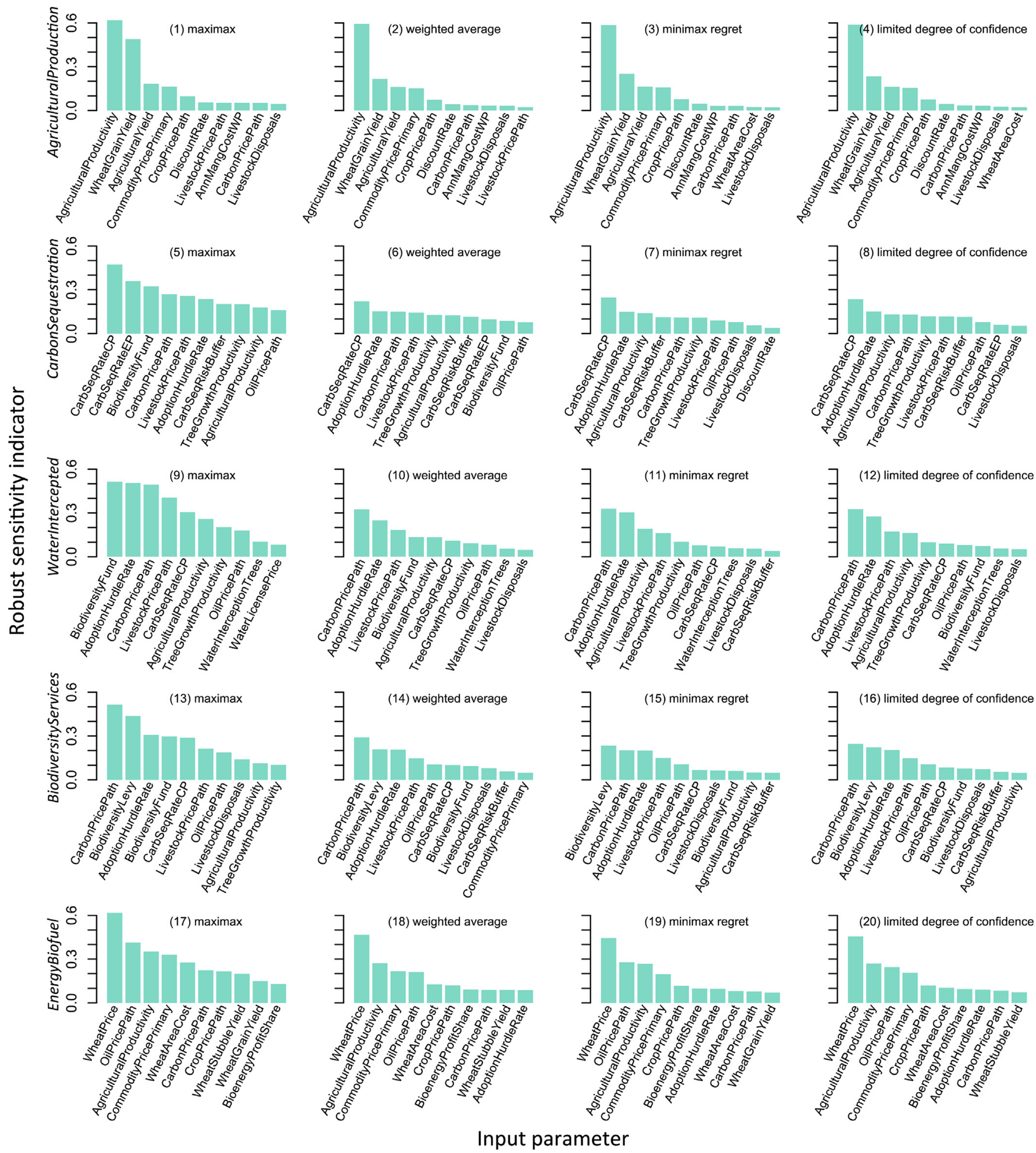

Fig. 5. Robust sensitivity indicators for the top ten most influential input parameters for the five key output variables using the four decision criteria.

change to the complex non-linear system and led to variation in uncertainty in model outputs generated from variation in parameter inputs. This is a novel and important contribution. Most scenario exercises (e.g., Millennium Ecosystem Assessment, 2005a; Moss et al., 2010; van Vuuren et al., 2011; Bateman et al., 2013; Howells et al., 2013), have been conducted to help decision makers understand a range of possible futures and create robust management strategies. We believe that none of these such exercises explored the scope and the distribution of model outcomes under different scenarios resulting from variation in model inputs.

The finding that significant differences exist in parameters identified as influential and non-influential, in parameter importance rankings, and in the magnitude of first-order and total effects under deep uncertainty, has not been reported previously. While several global sensitivity analyses have assessed model uncertainty and parameter sensitivity under future conditions (e.g., Anderson et al., 2014; Butler et al., 2014), none were cognisant of the influence of deep uncertainty on the results. These findings are important because they provide solid evidence supporting the need for robust GSA to guide further data collection and analyses when faced with deep uncertainty. Exploring the dependence of GSA results on model scenarios (deeply uncertain conditions under which models are run) is critical. Failure to do so could result in the targeting of parameters for modelling, data collection, and refinement that may not be influential under certain scenarios. Critically, it may also lead to the exclusion from these efforts of parameters that have been found to be non-influential, but which are highly influential under alternative unanalysed scenarios. It is likely that these findings are generalisable beyond the LUTO model and can be extrapolated to other social-ecological models that run under deep uncertainty.

Here, we focussed on the most influential parameters that need to be prioritised for future investment in modelling-related activities. A common alternate use of the same information is in the identification of non-influential parameters for the purpose of model simplification (Saltelli et al., 2006). For the LUTO model, few non-influential parameters were identified under each scenario and very different results were obtained between scenarios. The result was due to the inherent complexity of the LUTO model, external influences (expressed by scenarios) on the model, and the thresholds for identifying non-influential parameters. This result confirms the importance of considering deep uncertainty when the focus of GSA is on identifying non-influential model parameters (i.e. parameter screening).

5.2. Decision making for deeply uncertain futures

We needed to quantify sensitivity in a way that was robust to deep uncertainty and can be used to reliably inform investment in data collection, modelling, and refining of parameter estimates. We proposed four decision criteria to find robust sensitivity indices that work acceptably well under all scenarios. Following decision theory, we calculated four new sensitivity indicators based on the decision objective, which incorporated the decision maker's attitude to risk. No decision criterion is the *right* or *best* choice. Success in the future is based on a decision's future success, which we cannot know in advance. Our aim was not to make a judgement about which criterion should be applied by the decision maker, but rather to provide a number of options to support decision-making under deep uncertainty. A specific criterion can be selected based on the objective and level of risk aversion of the decision maker. The maximax criterion is 'aggressive' and suitable for the decision maker who is not willing to omit any possible influential parameter. The minimax regret criterion is relatively conservative and fits those who want to minimise the regret of the worst case—accepting sensitivity indices from a state of world that does not eventuate. The weighted average criterion is recommended to those who expect to convert 'deep uncertainty' to 'probabilistic uncertainty' and have their own confidence in the assessment for the probabilities of sensitivity indices from different scenarios. The LDC criterion is balance of weighted average and minimax regret, and the weighting in LDC can be explained as a measure of the decision maker's confidence in the occurrence probabilities of different future scenarios. Choice of decision criterion can make the decision maker comfortable with the ambiguity of an open future (Gao and Hailu, 2012, 2013). Robust sensitivity metrics allowed us to identify those parameters that are most influential for specific outputs irrespective of scenario. This ensures that focussing attention on the most robustly influential parameters will be most efficient under the kind of deep uncertainty characteristic of social-ecological systems.

Decisions must be made about how to cope with and take advantage of the significant differences that exist in uncertainty of outputs and sensitivity of inputs between scenarios. New, robust technical methods have recently emerged to contribute to the robust risk analysis and decision making in the presence of deep uncertainty, including model ensembles, adaptive management, robust optimisation, and decision algorithms (Cox, 2012). These methods usually employ multiple models and scenarios, and with these methods, the decision maker needs to act intelligently on the basis of possible future changes. When the degree of uncertainty is high and uncertainty is not controllable (for instance, unknown futures), scenario planning and decision algorithms are effective ways of coping (Peterson et al., 2003; Cox, 2012). Application of these and other robust technical methods and integrating them into a formal set of mathematical and computational processes for robust global sensitivity analysis seems a productive path for future research.

5.3. Limitations and future directions

There are two main limitations in our GSA experiments. First, to reduce the parameter dimensionality and computational demands, the variation of temporal and spatial parameters were simplified by introducing a multiplier ranging from 0.7 to 1.3 (i.e. $\pm$30%) uniformly applied to all time-series and spatial parameters. This ignored the variation characteristics and relations of parameters that may change non-uniformly over time and/or space. Some recent progress has been made in investigating the effect of spatial parameters on sensitivity (e.g., Dong et al., 2015) but more effort is required. In addition, these variation ranges could have a significant influence on the parameter sensitivity (Wang et al., 2013) and more effort to refine parameter boundary conditions of influential parameters would further improve the GSA results.

The second major limitation is the likelihood that the definition of the four global outlooks as scenarios was not sufficient to capture the range of future uncertainty in all parameters (Bryant and Lempert, 2010). Recently, a crucial contribution to decision making under deep uncertainty are bottom-up approaches of scenario discovery (Herman et al., 2015) which advocate the generation and analysis of many (e.g., thousands of) scenarios. Examples include Robust Decision Making (Groves and Lempert, 2007; Lempert and Collins, 2007; Bryant and Lempert, 2010; Kwakkel et al., 2014), Decision Scaling (Brown et al., 2012; Brown and Wilby, 2012), Info-Gap (Ben-Haim, 2004; Matrosov et al., 2013), and Many-Objective Robust Decision Making (Kasprzyk et al., 2013; Herman et al., 2014). One challenge with using scenario discovery techniques to incorporate many scenarios into sensitivity analysis of complex systems models is the sheer amount of effort involved in their specification. In our case, scenario parameters were quantified by integrated assessment to produce the input parameters shown in Table 1 which is in itself a highly technical and effortful task. Another challenge with incorporating many scenarios is that it adds significant computational load to global sensitivity analyses that are already pushing present-day computational limits (Song et al., 2012; Zhao et al., 2014). For example, for computational tractability in this study we used a coarse resolution version of the LUTO model (9.9 km^2 grid cells instead of 1.2 km^2 at the highest model resolution). Each model run took approximately 1–2 h, totalling at least 7250 h ($145 \times 50 \times 1$) to undertake the eFAST analysis under one scenario. Analysing hundreds or thousands of scenarios presents a significant computational challenge, even at coarse resolution and with access to high performance computing (Bryan, 2013). To overcome this barrier, we are currently investigating the potential of screening methods as robust GSA approaches with greatly reduced computational demand. We incorporated scenarios into the p-level sampling space (p is the number of levels to which each dimension of the parameter space is divided) of Morris' elementary effects method (Gao and Bryan, in press). However, as the choice of p is strictly linked to the choice of r (the number of trajectories sampled), the increase of scenarios inevitably adds computational cost. Further advances are required to overcome these technical and computational challenges and enable scenarios to capture a greater range of future uncertainty in sensitivity analyses of complex models via scenario discovery techniques, and thereby improve the robustness of produced sensitivity metrics.

6. Conclusion

Using the eFAST method, we produced comprehensive model diagnostics of a large, integrated social-ecological systems model under conditions of deep uncertainty characterised by four global change scenarios, each of which represents a set of plausible, internally consistent future conditions. We found significant differences between scenarios in model outputs that resulted from variation in model inputs. The contributions of input parameters to the scope and distribution of each model output were also highly scenario-dependent. The GSA results revealed that global scenarios had a significant effect on the influential and non-influential parameters, parameter importance ranking, and the magnitude of both first-order and total effects. Four criteria from decision theory were used to successfully integrate parameter sensitivity estimates across scenarios and provide options for decision makers to determine new sensitivity indicators that were robust to deeply uncertain conditions. The decision-maker can then evaluate the decision criteria against their own risk preferences and more robustly quantify sensitivity for reliably informing further investment in data collection, modelling, and refining of parameter estimates. These findings are of general value and can be extrapolated to other models that must run under deep uncertainty. The proposed methods are a valuable addition to global sensitivity analyses that can also increase our understanding of the effects of deep uncertainty on output uncertainty and parameter sensitivity, incorporate the decision maker's risk preference into modelling-related activities, and obtain greater resilience of decisions to surprise.

Acknowledgements

We are grateful for the support of CSIRO Agriculture, CSIRO Land and Water, and the Australian National Outlook initiative. We also appreciate the comments of three anonymous reviewers which have greatly improved this manuscript.

Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.envsoft.2015.11.001.

References

Anderson, B., Borgonovo, E., Galeotti, M., Roson, R., 2014. Uncertainty in climate change modeling: can global sensitivity analysis be of help? Risk Anal. 34 (2), 271–293.
Anderson, D.R., Sweeney, D.J., Williams, T.A., Camm, J.D., Martin, R.K., 2012. An Introduction to Management Science: quantitative Approaches to Decision Making. Cengage Learning, Mason, OH, USA.
Annoni, P., Brüggemann, R., Saltelli, A., 2011. Partial order investigation of multiple indicator systems using variance-based sensitivity analysis. Environ. Model. Softw. 26 (7), 950–958.
Bateman, I.J., Harwood, A.R., Mace, G.M., Watson, R.T., Abson, D.J., Andrews, B., Binner, A., Crowe, A., Day, B.H., Dugdale, S., 2013. Bringing ecosystem services into economic decision-making: land use in the United Kingdom. Science 341 (6141), 45–50.
Ben-Haim, Y., 2001. Info-Gap Decision Theory. Academic Press, San Diego, CA.
Ben-Haim, Y., 2004. Uncertainty, probability and information-gaps. Reliab. Eng. Syst. Saf. 85 (1–3), 249–266.
Bohensky, E.L., Reyers, B., Van Jaarsveld, A.S., 2006. Future ecosystem services in a southern African river basin: a scenario planning approach to uncertainty. Conserv. Biol. 20 (4), 1051–1061.
Brown, C., Ghile, Y., Laverty, M., Li, K., 2012. Decision scaling: linking bottom-up vulnerability analysis with climate projections in the water sector. Water Resour. Res. 48 (9), W09537.
Brown, C., Wilby, R.L., 2012. An alternate approach to assessing climate risks. Eos Trans. Am. Geophys. Union 93 (41), 401–402.
Bryan, B.A., 2013. High-performance computing tools for the integrated assessment and modelling of social–ecological systems. Environ. Model. Softw. 39, 295–303.
Bryan, B.A., Crossman, N.D., King, D., Meyer, W.S., 2011. Landscape futures analysis: assessing the impacts of environmental targets under alternative spatial policy options and future scenarios. Environ. Model. Softw. 26 (1), 83–91.
Bryan, B.A., Nolan, M., Harwood, T.D., Connor, J.D., Navarro-Garcia, J., King, D., Summers, D.M., Newth, D., Cai, Y., Grigg, N., Harman, I., Crossman, N.D., Grundy, M.J., Finnigan, J.J., Ferrier, S., Williams, K.J., Wilson, K.A., Law, E.A., Hatfield-Dodds, S., 2014. Supply of carbon sequestration and biodiversity services from Australia's agricultural land under global change. Glob. Environ. Change 28, 166–181.
Bryan, B.A., Nolan, M., McKellar, L., Connor, J.D., Newth, D., Harwood, T., King, D., Navarro, J., Cai, Y., Gao, L., Grundy, M., Graham, P., Ernst, A., Dunstall, S., Stock, F., Brinsmead, T., Harman, I., Grigg, N.J., Battaglia, M., Keating, B., Wonhas, A., Hatfield-Dodds, S., 2015. Land-use and sustainability under intersecting global change and domestic policy scenarios: trajectories for Australia to 2050. Global Environmental Change (in review).
Bryant, B.P., Lempert, R.J., 2010. Thinking inside the box: a participatory, computer-assisted approach to scenario discovery. Technol. Forecast. Soc. Change 77 (1), 34–49.
Butler, M.P., Reed, P.M., Fisher-Vanden, K., Keller, K., Wagener, T., 2014. Identifying parametric controls and dependencies in integrated assessment models using global sensitivity analysis. Environ. Model. Softw. 59, 10–29.
Chen, L., Tian, Y., Cao, C., Zhang, S., Zhang, S., 2012. Sensitivity and uncertainty analyses of an extended ASM3-SMP model describing membrane bioreactor operation. J. Membr. Sci. 389, 99–109.
Confalonieri, R., Bellocchi, G., Tarantola, S., Acutis, M., Donatelli, M., Genovese, G., 2010. Sensitivity analysis of the rice model WARM in Europe: exploring the effects of different locations, climates and methods of analysis on model sensitivity to crop parameters. Environ. Model. Softw. 25 (4), 479–488.
Connor, J.D., Bryan, B.A., Nolan, M., Stock, F., Gao, L., Dunstall, S., Graham, P., Ernst, A., Newth, D., Grundy, M., Hatfield-Dodds, S., 2015. Modelling Australian land use competition and ecosystem services with food price feedbacks at high spatial resolution. Environ. Model. Softw. 69, 141–154.
Corder, G.W., Foreman, D.I., 2009. Nonparametric Statistics for Non-statisticians: a Step-by-step Approach. Wiley, New Jersey, USA.
Cosenza, A., Mannina, G., Vanrolleghem, P.A., Neumann, M.B., 2013. Global sensitivity analysis in wastewater applications: a comprehensive comparison of different methods. Environ. Model. Softw. 49, 40–52.
Cosenza, A., Mannina, G., Vanrolleghem, P.A., Neumann, M.B., 2014. Variance-based sensitivity analysis for wastewater treatment plant modelling. Sci. Total Environ. 470–471, 1068–1077.
Cox, L.A., 2012. Confronting deep uncertainties in risk analysis. Risk Anal. 32 (10), 1607–1629.
Davies-Barnard, T., Valdes, P.J., Singarayer, J.S., Pacifico, F.M., Jones, C.D., 2014. Full effects of land use change in the representative concentration pathways. Environ. Res. Lett. 9 (11), 114014.
DeJonge, K.C., Ascough Ii, J.C., Ahmadi, M., Andales, A.A., Arabi, M., 2012. Global sensitivity and uncertainty analysis of a dynamic agroecosystem model under different irrigation treatments. Ecol. Model. 231, 113–125.
Dong, M., Bryan, B.A., Connor, J.D., Nolan, M., Gao, L., 2015. Land use mapping error introduces strongly-localised, scale-dependent uncertainty into land use and ecosystem services modelling. Ecosyst. Serv. 15, 63–74.
Freni, G., Mannina, G., 2010. Uncertainty in water quality modelling: the applicability of variance decomposition approach. J. Hydrol. 394 (3–4), 324–333.
Gan, Y., Duan, Q., Gong, W., Tong, C., Sun, Y., Chu, W., Ye, A., Miao, C., Di, Z., 2014. A comprehensive evaluation of various sensitivity analysis methods: a case study with a hydrological model. Environ. Model. Softw. 51, 269–285.
Gao, L., Bryan, B.A., 2015. Incorporating deep uncertainty into the elementary effects method for robust global sensitivity analysis. Ecol. Model. http://dx.doi.org/10.1016/j.ecolmodel.2015.10.016 (in press).
Gao, L., Connor, J.D, Dillon, P., 2014. The economics of groundwater replenishment for reliable urban water supply. Water 6 (6), 1662–1670.
Gao, L., Hailu, A., 2012. Ranking management strategies with complex outcomes: an AHP-fuzzy evaluation of recreational fishing using an integrated agent-based model of a coral reef ecosystem. Environ. Model. Softw. 31, 3–18.
Gao, L., Hailu, A., 2013. Identifying preferred management options: an integrated agent-based recreational fishing simulation model with an AHP-TOPSIS evaluation method. Ecol. Model. 249, 75–83.
Groves, D.G., Lempert, R.J., 2007. A new analytic method for finding policy-relevant

scenarios. Glob. Environ. Change 17 (1), 73–85.
Hatfield-Dodds, S., Brinsmead, T., Bryan, B., Graham, P., Grundy, M., Harwood, T., Newth, D., Schandl, H., Wonhas, A., Adams, P., McKellar, L.B., Cai, Y., Ferrier, S., Finnigan, J., Hanslow, K., McCallum, R., Nolan, M., Prosser, I., Smith, M.S., Baynes, T., Chiew, F., Connor, J., Geschke, A., Grigg, N., Harman, I., Hayward, J., Keating, B., King, D., Lenzen, M., Lonsdale, M., McCrae, R., Garcia, J.N., Owen, A., Raison, J., Reedman, L., Smith, M.H., Summers, D., Whetton, P., 2015. CSIRO Australian National Outlook 2014: Living Standards, Resource Use, Environmental Performance and Economic Activity, 1970-2050. CSIRO, Canberra.
Herman, J.D., Reed, P.M., Zeff, H.B., Characklis, G.W., 2015. How should robustness be defined for water systems planning under change? J. Water Resour. Plan. Manag. 141 (10), 04015012.
Herman, J.D., Zeff, H.B., Reed, P.M., Characklis, G.W., 2014. Beyond optimality: multistakeholder robustness tradeoffs for regional water portfolio planning under deep uncertainty. Water Resour. Res. 50 (10), 7692–7713.
Howells, M., Hermann, S., Welsch, M., Bazilian, M., Segerstrom, R., Alfstad, T., Gielen, D., Rogner, H., Fischer, G., van Velthuizen, H., Wiberg, D., Young, C., Roehrl, R.A., Mueller, A., Steduto, P., Ramma, I., 2013. Integrated analysis of climate change, land-use, energy and water strategies. Nat. Clim. Change 3 (7), 621–626.
IPCC, 2000. Special Report on Emissions Scenarios. A Special Report of Working Group III of Tiie Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, UK.
IPCC, 2007. In: Writing Team, Core, Pachauri, R.K., Reisinger, A. (Eds.), Climate Change 2007: Synthesis Report. Contribution of Working Groups I, II and III to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. IPCC, Geneva, Switzerland, p. 104.
Jackson, S.L., 2012. Research Methods and Statistics: a Critical Thinking Approach. Cengage Learning, Belmont, CA, USA.
Kasprzyk, J.R., Nataraj, S., Reed, P.M., Lempert, R.J., 2013. Many objective robust decision making for complex environmental systems undergoing change. Environ. Model. Softw. 42, 55–71.
Kirby, J.M., Connor, J., Ahmada, M.D., Gao, L., Mainuddin, M., 2014. Climate change and environmental water reallocation in the Murray-Darling Basin: impacts on flows, diversions and economic returns to irrigation. J. Hydrol. 518, 120–129.
Kirby, M., Connor, J., Ahmad, M.-u.D., Gao, L., Mainuddin, M., 2015. Irrigator and environmental water management adaptation to climate change and water reallocation in the Murray–Darling Basin. Water Econ. Policy 01 (03), 1550009.
Knight, F.H., 1921. Risk, Uncertainty, and Profit. Houghton Mifflin, Boston, MA.
Kwakkel, J., Haasnoot, M., Walker, W., 2014. Developing dynamic adaptive policy pathways: a computer-assisted approach for developing adaptive strategies for a deeply uncertain world. Clim. Change 1–14.
Lempert, R.J., Collins, M.T., 2007. Managing the risk of uncertain threshold responses: comparison of robust, optimum, and precautionary approaches. Risk Anal. 27 (4), 1009–1026.
Loucks, D.P., van Beek, E., Stedinger, J.R., Dijkman, J.P.M., Villars, M.T., 2005. Water Resources Systems Planning and Management: an Introduction to Methods, Models and Applications. UNESCO, Paris, France.
Mahmoud, M., Liu, Y., Hartmann, H., Stewart, S., Wagener, T., Semmens, D., Stewart, R., Gupta, H., Dominguez, D., Dominguez, F., Hulse, D., Letcher, R., Rashleigh, B., Smith, C., Street, R., Ticehurst, J., Twery, M., van Delden, H., Waldick, R., White, D., Winter, L., 2009. A formal framework for scenario development in support of environmental decision-making. Environ. Model. Softw. 24 (7), 798–808.
Makler-Pick, V., Gal, G., Gorfine, M., Hipsey, M.R., Carmel, Y., 2011. Sensitivity analysis for complex ecological models – a new approach. Environ. Model. Softw. 26 (2), 124–134.
Marino, S., Hogue, I.B., Ray, C.J., Kirschner, D.E., 2008. A methodology for performing global uncertainty and sensitivity analysis in systems biology. J. Theor. Biol. 254 (1), 178–196.
Matrosov, E.S., Woods, A.M., Harou, J.J., 2013. Robust decision making and Info-Gap decision theory for water resource system planning. J. Hydrol. 494 (0), 43–58.
McInerney, D., Lempert, R., Keller, K., 2012. What are robust strategies in the face of uncertain climate threshold responses? Clim. Change 112 (3–4), 547–568.
Miao, Z., Lathrop Jr., R.G., Xu, M., La Puma, I.P., Clark, K.L., Hom, J., Skowronski, N., Van Tuyl, S., 2011. Simulation and sensitivity analysis of carbon storage and fluxes in the New Jersey Pinelands. Environ. Model. Softw. 26 (9), 1112–1122.
Millennium Ecosystem Assessment, 2005a. Ecosystems and Human Well-being: Scenarios. Island Press, Washington, DC.
Millennium Ecosystem Assessment, 2005b. Ecosystems and Human Well-being: Scenarios, vol. 2. Island Press, Washington, DC.
Morgan, M.G., Henrion, M., 1992. Uncertainty: a Guide to Dealing with Uncertainty in Quantitative Risk and Policy Analysis. Cambridge University Press, Cambridge, UK.
Morgan, M.G., Henrion, M., Small, M., 1992. Uncertainty: a Guide to Dealing with Uncertainty in Quantitative Risk and Policy Analysis. Cambridge University Press, Cambridge, UK.
Moss, R.H., Edmonds, J.A., Hibbard, K.A., Manning, M.R., Rose, S.K., van Vuuren, D.P., Carter, T.R., Emori, S., Kainuma, M., Kram, T., Meehl, G.A., Mitchell, J.F.B., Nakicenovic, N., Riahi, K., Smith, S.J., Stouffer, R.J., Thomson, A.M., Weyant, J.P., Wilbanks, T.J., 2010. The next generation of scenarios for climate change research and assessment. Nature 463 (7282), 747–756.
Nossent, J., Elsen, P., Bauwens, W., 2011. Sobol' sensitivity analysis of a complex environmental model. Environ. Model. Softw. 26 (12), 1515–1525.
Peterson, G.D., Cumming, G.S., Carpenter, S.R., 2003. Scenario planning: a tool for conservation in an uncertain world. Conserv. Biol. 17 (2), 358–366.
Polasky, S., Carpenter, S.R., Folke, C., Keeler, B., 2011. Decision-making under great uncertainty: environmental management in an era of global change. Trends Ecol. Evol. 26 (8), 398–404.
Popper, S.W., Lempert, R.J., Bankes, S.C., 2005. Shaping the future. Sci. Am. 292, 66–71.
Quade, E.S., 1989. Analysis for Public Decisions, third ed. Elsevier Science, New York.
Saltelli, A., Annoni, P., Azzini, I., Campolongo, F., Ratto, M., Tarantola, S., 2010. Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index. Comput. Phys. Commun. 181 (2), 259–270.
Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., Tarantola, S., 2008. Global Sensitivity Analysis: the Primer. John Wiley & Sons, West Susses, England.
Saltelli, A., Ratto, M., Tarantola, S., Campolongo, F., 2006. Sensitivity analysis practices: strategies for model-based inference. Reliab. Eng. Syst. Saf. 91 (10–11), 1109–1125.
Saltelli, A., Tarantola, S., Chan, K.P.S., 1999. A quantitative model-Independent method for global sensitivity analysis of model output. Technometrics 41 (1), 39–56.
Schaldach, R., Alcamo, J., Koch, J., Kölking, C., Lapola, D.M., Schüngel, J., Priess, J.A., 2011. An integrated approach to modelling land-use change on continental and global scales. Environ. Model. Softw. 26 (8), 1041–1051.
Shin, M.-J., Guillaume, J.H.A., Croke, B.F.W., Jakeman, A.J., 2013. Addressing ten questions about conceptual rainfall–runoff models with global sensitivity analyses in R. J. Hydrol. 503, 135–152.
Sobol', I.M., 1993. Sensitivity analysis for nonlinear mathematical models. Math. Model. Comput. Exp. 1, 407–414.
Song, X., Bryan, B.A., Almeida, A.C., Paul, K.I., Zhao, G., Ren, Y., 2013. Time-dependent sensitivity of a process-based ecological model. Ecol. Model. 265, 114–123.
Song, X., Bryan, B.A., Paul, K.I., Zhao, G., 2012. Variance-based sensitivity analysis of a forest growth model. Ecol. Model. 247, 135–143.
Ullman, D.G., 2006. Making Robust Decisions: Decision Management for Technical, Business, & Service Teams. Trafford Publishing, Victoria, BC, Canada.
Van Oijen, M., Rougier, J., Smith, R., 2005. Bayesian calibration of process-based forest models: bridging the gap between models and data. Tree Physiol. 25 (7), 915–927.
van Rossum and The Python Community, 2013. The Python Programming Language: Version 2.7.5. The Python Software Foundation.
van Vuuren, D., Edmonds, J., Kainuma, M., Riahi, K., Thomson, A., Hibbard, K., Hurtt, G., Kram, T., Krey, V., Lamarque, J.-F., Masui, T., Meinshausen, M., Nakicenovic, N., Smith, S., Rose, S., 2011. The representative concentration pathways: an overview. Clim. Change 109 (1–2), 5–31.
Vanuytrecht, E., Raes, D., Willems, P., 2014. Global sensitivity analysis of yield output from the water productivity model. Environ. Model. Softw. 51, 323–332.
Varella, H., Guérif, M., Buis, S., 2010. Global sensitivity analysis measures the quality of parameter estimation: the case of soil parameters and a crop model. Environ. Model. Softw. 25 (3), 310–319.
Vezzaro, L., Mikkelsen, P.S., 2012. Application of global sensitivity analysis and uncertainty quantification in dynamic modelling of micropollutants in stormwater runoff. Environ. Model. Softw. 27–28, 40–51.
Walker, J., Almond, P., 2010. Interpreting Statistical Findings. McGraw-Hill Education, Berkshire, England.
Walker, W.E., Marchau, V.A.W.J., Swanson, D., 2010. Addressing deep uncertainty using adaptive policies: Introduction to section 2. Technol. Forecast. Soc. Change 77 (6), 917–923.
Wang, J., Li, X., Lu, L., Fang, F., 2013. Parameter sensitivity analysis of crop growth models based on the extended Fourier Amplitude Sensitivity Test method. Environ. Model. Softw. 48, 171–182.
Wilkinson, A., Kupers, R., 2013. Living in the futures. Harv. Bus. Rev. 91 (5), 119–127.
Wintle, B.A., Runge, M.C., Bekessy, S.A., 2010. Allocating monitoring effort in the face of unknown unknowns. Ecol. Lett. 13 (11), 1325–1337.
Zhao, G., Bryan, B.A., Song, X., 2014. Sensitivity and uncertainty analysis of the APSIM-wheat model: interactions between cultivar, environmental, and management parameters. Ecol. Model. 279, 1–11.